Manabot: What We Have Demonstrated and the Recipes We Use

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Written by AI. Not yet intended for distribution.

1 What we have demonstrated

Manabot trains agents to play a bounded, two-player subset of *Magic: The Gathering*. We now have repeated full-game learning across ten authored decks, a four-seed comparison of one-seat and two-seat learning, and a direct specialist-versus-generalist comparison. Stronger opponents expose weaknesses hidden by the earlier anchor. A separate pretrained-language-model proof completes one small native game; it is an integration result, not a strength benchmark.

This is a selective account of retained results through October 26, not a new experiment or pooled meta-analysis. Each number retains its world, recipe, seeds, opponents and cost basis. The Elemental studies use w13 and historical additive policies, not the current FiLM/policy-input-v2 recipe. Reports stored outside git are digest-indexed in the artifact manifest; `paper/README.md` gives exact fetch commands. Section 2 describes current source separately.

1.1 Three seeds learned across the ten-deck Elemental roster

The completed MTG-181 trial trained three seeds of one two-seat self-play recipe: a width-64, two-layer, four-head semantic policy with a scalar masked-mean critic, 256 streams, uniform ordered deck-pair sampling including mirrors, and damped policy learning. Each L4 allocation lasted five hours. Endpoints reached 474–616 updates and 22.4–27.4 million learner decisions; inclusive conservative rental cost was approximately $9.12 of a $10 cap, with cleanup confirmed.

Each scored checkpoint played a frozen 10-by-10 grid against heuristic v1: 50 games per cell, 5,000 per grid. All 13 grids completed with zero invalid games and endpoint load, legality, completion and replay checks passed. Mean overall score rose from 20.6% at initialization to 85.1% at the endpoints. Every deck improved. Endpoint spread across seeds was 1.5 percentage points; the predicted 35–70% endpoint band was refuted high.

An explicitly separate, post-protocol addendum scored the same endpoints against the strengthened heuristic, on the same grid. All 15,000 games were valid. The mean fell to 60.2% (seed standard deviation 2.6 points), and the seed range widened to 4.8 points:

Seed	Endpoint update	Heuristic v1	Strengthened heuristic
18101	616	85.1%	58.4% [57.0, 59.8]
18102	474	84.4%	59.0% [57.6, 60.3]
18103	568	85.9%	63.2% [61.9, 64.6]

These are reported 95% Wilson game intervals, not training-seed intervals or paired-deal bootstrap intervals. The newer anchor separates endpoints that v1 barely distinguished. UW is the policy's only losing deck on average against it (37.3%); UR is best (75.1%). Training length alone does not explain the seed ordering: seed 18102's shorter run slightly outscored 18101. This establishes repeated learning against declared anchors in one authored world, not broad *Magic* competence.

Failures remain part of the result. The earlier w10 MTG-174 attempt stopped after four updates on illegal mana activation, retaining initialization but no trained checkpoint. MTG-181's seed 18102 instead stopped at its allocation deadline: all 256 unfinished games were deadline cuts, despite the old record labeling them a game-error-rate stop. No engine or agent game failed in that trial. Those two outcomes must not be conflated.

1.2 Learning from both seats reduced collection cost

The completed from-scratch seat comparison used four seeds per arm on Allies/Lessons, 256 streams, complete games, identical initial weights within seed, and matched learner-decision milestones through 64 million. The one-seat arm collected two games per stream per update; the two-seat arm collected one and learned from both players. All eight runs passed 64 million learner decisions and all 60 scheduled evaluations completed.

At 8 million learner decisions, mean heuristic scores were 53.0% for one seat and 57.6% for both. At 64 million they were 67.1% and 65.9%; seed standard deviations were 2–6 points. The small early advantages do not establish a reliable method-level strength advantage or statistical equivalence. At matched learner decisions, two-seat training needed about half the games and reached 64 million decisions in a mean 7.81 training hours versus 10.74, approximately 1.4 times faster. The saving came from collection; learning time per learner decision was unchanged. Inclusive cost was about $54.40, including calibration and an earlier stopped warm-start attempt. Setup is excluded from the training-hour comparison.

At matched environment decisions two-seat led in all 19 reported seed comparisons, but the one-seat values were interpolated in log environment decisions. That is a descriptive comparison, not an additional independently evaluated cohort. The operational case for two-seat collection is stronger than the evidence for better learning per retained row.

1.3 A specialist gained narrowly on its matchup and lost elsewhere

MTG-186 trained one UR-versus-GW-only seed and compared it directly with MTG-181 seed 18101, at updates 200, 400, 600 and their common endpoint of 616. Each pairing played 1,000 UR/GW games balanced over decks and seats, plus 1,000 games sampled from the other 98 ordered pairs. All 8,000 games were valid and replayed. The specialist's UR/GW scores were 58.4%, 56.8%, 51.9% and 55.4%: the margin did not grow with training, and the update-600 interval included 50%.

At the endpoint, the specialist scored 55.4% [52.3, 58.5] on its matchup and 37.5% [34.6, 40.5] elsewhere (reported 95% Wilson intervals). A further 2,000 valid endpoint games on 500 fresh deals scored 56.2% [54.0, 58.4]. This is one seed on each side; the game intervals do not estimate seed variance, and shared-deal dependence is not captured by these Wilson summaries. Equal updates also did not mean equal data or compute: the specialist retained 20.49 million decisions versus 26.88 million for the generalist, with different collection/learning times. Inclusive settled cost was $3.72, covering failed launches, training, cancelled evaluation and scored cohorts.

Deck asymmetry matters. In the fresh-deal cohort, the specialist scored 83.9% as UR and 28.5% as GW. The model comparison requires both deck assignments; the high UR number alone mostly

measures the matchup. Associated Games-in-Hand card tables also mix deck strength with card occurrence. They describe associations, not causal card values or a mechanism for the specialist's gain.

1.4 The judge and the interface need their own measurements

The heuristic revision study used one MTG-183 checkpoint as opponent. On 300 held-out UR-versus-GW games, revising timing, targeting, blocking and Lesson choices raised the heuristic from 18.3% to 62.0%. A separate fixed-game single-process timing probe fell from 27.0 to 23.4 microseconds per decision. This changed anchor must not be spliced into v1 learning curves. Flat determinized search at 16 worlds reached 68.3%, a paired-deal gain of 6.3 points $[0.0, 12.7]$, at 346 milliseconds per decision under six concurrent games. The point estimate improved at a large cost; the interval touches zero.

The cached Qwen3.5-9B GPU proof won one seed-23 functional game against random on an L4: a 60-card mirror containing only Mountains, Gray Ogres and Lightning Bolts. All 160 model selections were valid on the first response, with no repairs or fallbacks; all 308 game Commands replayed on the worker and locally. Game time was 280.55 seconds, median model decision 1.96 seconds, loading 10.04 seconds, and the largest prompt 4,329 tokens. Sampled device memory peaked at 22,494 of 23,034 MiB. This establishes fit and completion for that one game at its original renderer/source, not general 8,192-token capacity, later-renderer performance, or playing strength.

1.5 Earlier negative results still bound our claims

The October batching study made the 64-by-2 recipe cheaper at fixed experience: 109–111 minutes at four streams versus 4.8–5.2 at 256 on RTX 5090. Optimizer-step counts changed too, so this is a whole-recipe result. More optimization did not reliably improve play: the three-seed advantage-floor screen gave a value-token floor-removal effect of -1.00 points $[-11.33, 10.33]$; the critic-coverage screen cost 45% more wall time for an uncertain $+3.33$ points $[-4.33, 11.33]$. These paired seed/deal intervals do not prove equivalence.

The July reset-seeding bug invalidated extreme random-play seat claims: repeated resets could replay one opening deal. Corrections F1–F3 retain the retraction and corrected mirror parity. A failed expert-iteration crank and known-correct control scenarios further showed that attractive aggregate wins need not imply competent decisions. Their earlier worlds and limitations remain attached to them.

What remains unestablished. No result here supplies held-out human strength, unseen-card transfer, a general exploitability bound, or an advantage over PPO. MTG-183's longer-training protocol and rating machinery are not a completed saturation result: no final report is retained in this source snapshot. The retained reports, rather than live progress or an expected endpoint, define what is demonstrated.

2 Methods: recipes and their influences

This section assumes familiarity with RL and describes implementable choices rather than the general difficulty of card games. Code paths refer to repository base `558e0186b` (October 26). Experiments above retain their own source identities. The recipes combine established methods with local engineering choices; they are not proposed as new algorithms.

2.1 Keep rules, observations and experiment identity together

Recipe. Use managym as the authoritative Rust rules engine and manabot as its learning client. Construct the authored deck and sideboard setup, encode only the acting viewer's observation, and score the engine's legal offers. Bind checkpoints to native world version, rules schemas, both deck/sideboard pairs, content manifest and input schema. Reject mismatches before loading weights. A rules-meaning or observation/action-shape change starts a new world; never draw a single learning curve across incompatible worlds.

Implementation. `manabot/env/` owns wrappers and encoding; `manabot/model/world.py` owns checkpoint bindings. Academy is the inspection/play client and does not define the training objective. Etude is the separate experiment launching/recording system. The deterministic, forkable engine makes simulated branches practical, but a sampled hidden world is not an observation that a policy may inspect. Source-pinned replay and viewer-equivalence checks are local integrity requirements, not borrowed claims of strategic correctness.

Current development is w13. Fresh observation configurations select policy input v2; saved configurations missing a version retain v1. The shared native decision capture supplies current stack and pending-payment relations to tensor and text clients. The v2 encoder uses a bounded 64-by-32 relation table with frame order, parent links, targets, costs and visible object/definition joins; overflow rejects input. `manabot/model/continuation.py` projects this stateless context into object and action representations. Optional public history is separately versioned. Neither current support nor a representation fix retroactively changes the inputs of the measured Elemental policies.

2.2 Score legal offers from objects and state

Recipe. Embed visible players, cards and permanents; apply masked object attention with perspective/ownership information; gather each legal action's type/features and its referenced object embeddings; produce one logit per offer and mask illegal padding. The compiled-match path also embeds checked semantic card programs. These joins distinguish card definitions, but their existence is not evidence of zero-shot transfer.

The attention block follows the multi-head attention/residual/normalization pattern of Vaswani et al. [2017]. It is a small object encoder, not the original sequence-to-sequence Transformer. New-run policies additionally use a scale-only adaptation of FiLM [Perez et al., 2018]. With encoded objects h_j, validity m_j, and informed action embedding e_a:

$$s = \frac{\sum_j m_j h_j}{\max(1, \sum_j m_j)}, \qquad \widetilde{e}_a = e_a \odot (1 + g(s)).$$

A shared head scores $\widetilde{e}_a$. Unlike a state-only additive offset shared by every logit, this multiplication can change relative action preferences. There is no separate FiLM shift term here. The masked mean and shared action head preserve object/offer permutation contracts.

Implementation. `ActorCritic` in `manabot/model/actor_critic.py` owns object attention, focus gathering, FiLM and the critic readout. The current ordinary baseline uses a width-64, two-layer, four-head semantic model with a scalar masked-mean critic and `policy_film=True`. Omitted

FiLM settings retain historical additive-policy semantics, including MTG-181/186. Value-token and categorical-outcome models are explicit alternatives; do not relabel a scalar checkpoint as categorical or attribute earlier additive-policy results to FiLM or policy input v2.

2.3 Collect transitions between decisions of the same player

Recipe. Batch independent games and freeze collection weights within each batch. Default self-play learns from both seats: collect at least the declared combined decision floor per stream, then finish current games before updating. Pack each player's trajectory separately, with end markers between players and games. Opponent actions are not the next step of the other player's return. One-seat exact-quota controls may span updates; explicit games-per-stream controls instead finish the stated number of games. Save actions, full legal behavior log probabilities, values and terminal outcomes. Bootstrap incomplete tails only from the exact paused next observation. Treat truncation as failure, not as a draw.

Influences and implementation. The PPO control uses Schulman et al. [2017] and GAE [Schulman et al., 2016]. The earlier trainer followed CleanRL's flat rollout-tensor organization [CleanRL contributors, 2026]; the maintained path now shares collection and learner ownership across objectives. `SeatRoutedCollector` in `training/rollout/collection.py` and `transition_gae` in `training/learning/returns.py`, both under `manabot/`, supply same-viewer credit. For transition-end flag d_t:

$$\delta_t = r_t + \gamma(1 - d_t)V_{t+1} - V_t, \qquad A_t = \delta_t + \gamma\lambda(1 - d_t)A_{t+1}.$$

Targets and saved behavior estimates are detached. Terminal signed outcomes label training rows without exposing true hidden cards as policy inputs. The one-seat path spanning updates is close-to-on-policy continuation; there is no trajectory-level correction for changing historical state occupancy. `Learner` owns train mode and Adam; `execute_regime` owns collection, update clocks, EMA, exports and recovery. Learning from both seats doubles eligible rows per game, not necessarily speed or statistical information.

2.4 Use the damped move update as an explicit adaptation

Influence. The ordinary self-play recipe adapts Ataraxos's move learning, specifically supplement S3.4, equations (5)–(6), and Table S7 [Sokota et al., 2026], including its clipped likelihood-ratio surrogate. It does not implement the full Stratego system or DeepNash's R-NaD.

Recipe. With current policy π, saved collection policy μ, detached advantage A, reference ρ and selected-action ratio $r = \pi(a)/\mu(a)$, minimize the selected-row mean of

$$-\min\{rA, \mathrm{clip}(r, 1-\epsilon, 1+\epsilon)A\} + c\, D_{\mathrm{KL}}(\pi\|\mu) + \tau\, D_{\mathrm{KL}}(\pi\|\rho).$$

Both reverse KLs sum over the entire legal support. The first damps movement from collection behavior; the second pulls toward a fixed reference. Our default reference chooses uniformly among present action types, then uniformly among offers of the chosen type. This replaces Stratego's piece/move grouping with an MTG engineering choice; a spell with many targets does not receive proportionally more prior mass solely from offer count.

Use $\gamma = 1$, policy trace $\lambda = .5$, and value trace $\lambda = .8$. Keep rows whose absolute advantage reaches the inclusive .75 quantile and the .01 floor; ties can retain more than one quarter. Do not normalize advantages after selection. Default filtering applies to actor and critic; the actor-only switch keeps all critic rows. Perform one epoch in collection order, without shuffling. Complete-game packing puts consecutive same-player trajectories into one column; the native stream count

sets the minibatch width. Earlier minibatches change parameters, so saved action likelihoods still matter within that epoch.

For categorical loss/draw/win heads, terminal targets are one-hot outcomes and nonterminal targets recursively mix next predictions and lambda returns; minimize soft-target cross-entropy. The ordinary baseline instead uses a scalar critic and MSE lambda-return loss, an explicit representation departure from Ataraxos. The value loss has unit weight. Gradients reach current policy/critic heads and their shared encoder; behavior distributions, references and targets remain fixed.

Implementation and constants. Under `manabot/training/learning/`, `damped_policy.py` owns loss, targets and update grouping, `references.py` the reference, and `selection.py` selection. `DampedPolicyLearning` in `training/regime.py` sets $\epsilon = .2$, $c = .1$, gradient norm cap .267, and

$$\eta_i = \text{clip}(.5/i^{1.1}, 5\times10^{-6}, 10^{-4}), \qquad \tau_i = .05/i^{.3}.$$

Adam uses the existing betas .9/.999 and epsilon 10^{-5}. Evaluation EMA uses decay .999, updated once per collection/update iteration, including empty filters; it is not a fictitious-play mixture. The source leaves some details unspecified (including Adam constants, quantile interpolation and EMA initialization); `docs/ataraxos.md` records our choices rather than claiming exact reproduction.

2.5 Freeze the whole recipe, including its clock

Recipe. Declare model, world/setup, learner, schedules, opponent mode, collection budget, checkpoint selection and evaluation before launch. Preserve raw and EMA exports separately. Collection updates, learner transitions, selected exposures and elapsed cost are distinct quantities. Retain empty filters and failed attempts in that accounting.

Implementation. `manabot/recipes/baseline.py` resolves `baseline-v3`: four streams with a floor of 64 combined decisions each, complete games from both seats, 128 updates by default, current-self collection, terminal outcome reward and the FiLM scalar model above. Its horizon-dependent schedule maps first and last positions onto historical iterations 1 and 10,000; a shorter run is a compressed schedule, not a prefix. The frozen Elemental regimes instead retain additive policies, 256 streams and absolute-iteration schedules. MTG-181 samples uniform ordered pairs from ten decks at game roots; MTG-186 fixes UR/GW. Uniform roots do not promise equal transition exposure. Neither recipe silently inherits the other's evidence.

PPO remains a selectable control, with its own discounting, normalization and reuse settings. Do not call a bundled PPO-versus-Ataraxos recipe comparison a single-mechanism ablation. Experiment modules resolve recipes; `TrainingRegime` and `TrainingRun` retain configuration, source/native/setup identities, seeds, exports and counters, with Etude owning current remote jobs, metrics and storage. Segmented learner continuation retains raw/EMA weights, Adam and absolute counters, but restarts collector/game/RNG streams. This is not byte-identical process recovery.

Opt-in game-error recovery quarantines failed games before optimizing their rows and stops above the declared rolling error threshold. Deadline-cut games are recorded as interruptions, not engine errors. Evaluation remains strict: missing or invalid games stay visible rather than being replaced to complete a favorable denominator.

2.6 Separate search labels from strategic evidence

Recipe. A frozen-policy local teacher samples compatible hidden worlds, evaluates every positive-support root action with frozen-policy continuations, and aggregates action values before choosing a

distribution. Every simulated actor uses its own ordinary observation. Compatible-prior sampling and a learned posterior are different declared choices. Exact-history likelihood support is restricted; unsupported events fail rather than becoming fabricated posterior evidence.

The local policy update follows Ataraxos supplement S3.7, equations (7)–(8). For base policy p, reference ρ, and sampled values $\widehat{Q}$, it maximizes

$$\sum_a q_a \widehat{Q}_a - \alpha D_{\mathrm{KL}}(q\|\rho) - \beta D_{\mathrm{KL}}(q\|p), \quad q_a \propto \exp\left(\frac{\widehat{Q}_a + \alpha \log \rho_a + \beta \log p_a}{\alpha + \beta}\right).$$

`manabot/sim/search/local_update.py` owns the teacher and receipts. This instrument is distinct from the July flat-search/distillation experiment. Supervised policy training has since been retired from the maintained runtime; historical distillation results require their pinned source. The maintained heuristic-search comparison above uses full heuristic rollouts, not this regularized local update or a learned belief posterior. Its measured gain cannot be transferred to these separate instruments.

2.7 Give text agents the same decision authority

Recipe. Render a viewer-safe native decision capture with the complete legal reference list; resolve the selected reference back to its revision-bound Command. The pretrained LLM harness uses pinned model/tokenizer revisions and chat templates, greedy BF16 text inference, and no fine-tuning or writable model memory. Accept exactly one legal reference completed by EOS; allow one repair for malformed output, then an explicitly marked seeded-uniform fallback. Runtime, context and deadline failures stop rather than becoming fallback choices. Count first-response validity, repairs and fallbacks separately.

Implementation. `manabot/llm/` owns selection and persistent model loading. Under `manabot/verify/`, `llm_game.py` binds the text renderer, Commands and replay. The game recipe declares bounded public history and includes full chat-template tokens in admission, without clipping. Reuse model weights between decisions, but retain no conversational/KV state across them. The October 25 GPU measurement predates the final capture-v2 renderer, so current interface support has no inherited latency, capacity or strength result.

2.8 Measure outcomes, behavior and cost separately

Recipe. Freeze an opponent cohort, independent training seeds, held-out paired deals, deck/seat assignments and inference envelope. Score complete games and replay every retained command. Keep inspected monitoring deals separate from a final holdout. Bootstrap whole paired-deal blocks for a fixed checkpoint; resample training seeds as well when estimating a method contrast. A large number of games from one seed cannot supply independent training replicates. Keep payoff matrices and competency failures alongside aggregate scores.

Implementation. `manabot/sim/` executes games and replay; `manabot/eval/` owns benchmark admission, monitoring and ratings. Etude experiment modules declare schedules, deliver game tables and render reports. The Elemental grid and specialist reports use game-level Wilson intervals; their shared deals and training seeds require separate uncertainty treatment for a method claim. Ratings are local to a world, matchup suite, information boundary, compute class and evaluator version. The old standalone learnability verifier is retired. Frozen reports, artifact manifests and replay receipts remain the authority for measurements, including failed attempts, missing evidence and cost exclusions.

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